

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-blind, Placebo-controlled, Multicenter Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab as Monotherapy or in Addition to Baseline Therapy in Patients With Myelin Oligodendrocyte Glycoprotein Antibody-associated Disease (MOGAD) **Short Title/Acronym:** Meteoroid **Protocol Number:** WN43194 (2023-507196-22-00) **NCT Number:** NCT05271409 **Posting ID:** 589755412928260862 **Trial Status:** Recruiting **Sponsor Name:** Hoffmann-La Roche / Roche **Investigational Product:** Satralizumab **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Trial Leader

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of satralizumab compared with placebo based on the time from randomization to the first occurrence of an adjudicated MOGAD relapse in the double-blind (DB) treatment period, as determined by an Adjudication Committee (CEC). **Secondary Objectives:** To evaluate the annualized rate of adjudicated MOGAD relapses, the annualized rate of active lesions on Magnetic Resonance Imaging (MRI) of the neuroaxis, the proportion of participants receiving rescue therapy, and to assess the safety, pharmacokinetics, and pharmacodynamics of satralizumab.

2.2 Background & Rationale To address the critical unmet medical need for targeted therapies in patients with Myelin Oligodendrocyte Glycoprotein Antibody-associated Disease (MOGAD). MOGAD is a rare, antibody-mediated inflammatory disease of the central nervous system that causes severe relapses leading to vision loss, muscle weakness, and neurological damage. Satralizumab, an IL-6 receptor inhibitor, is hypothesized to effectively prevent or delay these autoimmune attacks and reduce disease activity.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (with or without background immunosuppressant therapy) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 152 adolescents and adults with MOGAD (Event-driven sample size designed to reach a targeted number of relapses) **Number of Sites:** Multicenter

(Global) **Estimated Study Start:** August 2022 **Estimated Primary Completion:** November 2025 *Note: After the primary clinical cutoff date (CCOD), additional adolescent participants may be enrolled directly into the Open-Label Extension (OLE) period.*

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants who are aged ≥ 12 years at the time of signing the Informed Consent Form. 2. Confirmed diagnosis of MOGAD with a history of ≥ 1 MOGAD relapse in the 12 months prior to screening or ≥ 2 attacks in the 24 months prior to screening. 3. Expanded Disability Status Scale (EDSS) score of 0-6.5 at screening and High-contrast visual acuity (HCVA) better than 20/800 in each eye at screening. 4. Participants receiving either no or ongoing chronic immunosuppressant treatment (IST) for MOGAD at the time of screening. 5. For women of childbearing potential: participants who agree to remain abstinent or use adequate contraception during the treatment period and for at least 3 months after the final dose of satralizumab.

4.2 Exclusion Criteria 1. Presence of aquaporin-4-antibodies immunoglobulin G (AQP4-IgG) in the serum. 2. History of anti-N-methyl-D-aspartate receptor (NMDAR) encephalitis. 3. Any concomitant disease other than MOGAD that may require treatment with ISTs or oral corticosteroids (OCS) or intravenous (IV) corticosteroids at doses > 20 milligrams (mg) prednisone equivalent per day for > 21 days during the study. 4. Participants who are pregnant or breastfeeding, or intending to become pregnant during the study or within 3 months after the final dose of satralizumab. 5. Participants with active or presence of recurrent bacterial, viral, fungal, mycobacterial infection, or other infection at baseline. 6. Participants with evidence of latent or active tuberculosis (excluding patients receiving chemoprophylaxis for latent tuberculosis infection). 7. Participants with positive screening tests for hepatitis B and C. 8. Receipt of live or live-attenuated vaccine within 6 weeks prior to baseline. 9. History of severe allergic reaction to a biologic agent.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Satralizumab or matching placebo loading doses administered at Weeks 0, 2, and 4, followed by maintenance doses given every 4 weeks (Q4W) thereafter. **Route of Administration:** Subcutaneous (SC) injection in the abdominal or femoral region. **Duration of Treatment:** Up to 44 months in the double-blind (DB) treatment period, followed by an Open-Label Extension (OLE) period for up to 2 additional years where all participants receive satralizumab.

5.2 Concomitant & Prohibited Therapy Permitted: Baseline/ongoing chronic immunosuppressant treatments (ISTs) are permitted.
Prohibited: Concomitant diseases requiring ISTs, oral corticosteroids (OCS), or intravenous (IV) corticosteroids at doses > 20 mg prednisone equivalent per day for > 21 days during the study. Receipt of live or live-attenuated vaccines within 6 weeks prior to baseline is also prohibited.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, EDSS, MRI, HCVA, MOG-IgG/AQP4-IgG serology labs, Hepatitis B & C, TB screening
Baseline	Day 0	Randomization, First SC Dose of Satralizumab or Placebo
Interim	Weeks 2, 4, & Q4W	SC maintenance doses, safety checks, monitoring for relapse symptoms. (Option for at-home administration after 6 months)
End of DB Study	Up to Month 44	End of DB phase, MRI assessments, transition into Open-Label Extension (OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time from randomization to the first occurrence of a MOGAD relapse during the double-blind (DB) treatment period. **Measurement Method:** Clinical assessment determined and confirmed by an independent relapse Adjudication Committee (CEC).

7.2 Secondary & Exploratory Endpoints 1. Annualized Rate of Adjudicated MOGAD Relapses. 2. Annualized Rate of Active Lesions on Magnetic Resonance Imaging (MRI) of the Neuroaxis. 3. Proportion of Participants Receiving Rescue Therapy.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring including tracking for severe allergic reactions, injection site reactions, and potential opportunistic infections (bacterial, viral, fungal, mycobacterial). **Serious Adverse Events (SAEs):** Evaluated throughout the 44-month DB period and subsequent OLE period. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee reviewing safety signals and overseeing event-driven efficacy checkpoints.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy endpoints. Safety Set for AE/SAE monitoring. **Sample Size**

Justification: The trial is event-driven; participants are treated until the study records a predefined, statistically powered number of adjudicated MOGAD relapses across the treatment arms. Targeted enrollment is 152 participants. **Handling of Missing Data:** Standard survival analysis techniques (e.g., Kaplan-Meier curves, Cox proportional hazard models) to evaluate the time-to-first-relapse, accounting for right-censored data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite clinical check-ups for the first 6 months, followed by either continued onsite visits or home-based mobile nurse visits/self-administration. **Audit Trail:** Centralized reading for MRI scans and independent, blinded Adjudication Committee (CEC) for all reported relapse events.

11. Study Identifiers & Governance

Primary ID: WN43194 **Registry Number:** NCT05271409 **Posting ID:** 589755412928260862 **Sponsor/Lead Organization:** Hoffmann-La Roche / Roche **Study Title:** Meteoroid **Official Regulatory Title:** A Phase III, Randomized, Double-blind, Placebo-controlled, Multicenter Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab as Monotherapy or in Addition to Baseline Therapy in Patients With Myelin Oligodendrocyte Glycoprotein Antibody-associated Disease (MOGAD)

12. Clinical Framework (Disease Specific)

Primary Condition: Myelin Oligodendrocyte Glycoprotein Antibody-Associated Disease (MOGAD) **Diagnosis Threshold:** Confirmed MOGAD with ≥ 1 relapse in the prior 12 months or ≥ 2 attacks in the 24 months prior to screening **Medical Methodology:** Satralizumab (IL-6 receptor inhibitor) as monotherapy or with background immunosuppressant therapy (IST) **Optimization Target:** Prevention or delay of time to first MOGAD relapse

13. Study Procedures & Methodology

Management Pathway: Double-blind event-driven therapy followed by an Open-Label Extension (OLE) **Education Protocol (HCP):** Protocol-specific relapse evaluation criteria and MRI standardization **Education Protocol (Patient):** Subcutaneous injection training for at-home administration (after 6 months in-clinic) **Quality Audit Mechanism:** Independent Relapse Adjudication Committee (CEC) & Centralized MRI reads

14. Observation & Follow-Up Schedule

Total Duration: Up to 44 months DB Phase + up to 2 years OLE
Onsite Visit Cadence: Weeks 0, 2, 4, and every 4 weeks (Q4W)
initially **Remote Monitoring Frequency:** Continuous reporting for
suspected relapses; option for mobile nurse home administration
after 6 months **Checklist Review Interval:** Every 4 weeks (Q4W)
alongside maintenance dosing

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				